D02 · Demand Planning & Forecasting

04 · NPI demand forecasting

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1 WHAT IT IS

NPI demand forecasting builds a launch curve without history — using diffusion models, weighted structured analogs from comparable launches, and explicit adoption scenarios — to shape the ramp and its range, instead of copying a dissimilar predecessor or assuming a flat run-rate.

2 WHY IT MATTERS

Bass 1969 **(Bass, 1969 — peer-reviewed)** modeled new-product adoption as a mix of innovators and imitators, producing the characteristic launch ramp — a rise to a peak followed by decline — that a flat forecast cannot reproduce. A launch forecast by analogy to a poorly matched predecessor gets the ramp shape wrong: too steep means a launch stockout that hands early shelf and share to competitors during the one window when the category is undefended; too shallow means a write-off of obsolete launch inventory. NPI error of **40%+** at launch is normal, and the asymmetric cost — lost share versus markdown — makes the shape matter more than the level.

3 HOW IT IS DONE

1. **Build an analog library, not a memory.** Tag past launches by category, price tier, channel, region, and seasonality, and store their actual ramp curves — not just totals. The shape is the asset; the total is the byproduct.
2. **Select analogs by weighted similarity.** Three matched analogs, weighted by closeness, beat one planner's "it's like the last one." Document the match logic so the forecast is auditable, not a hunch wearing a number.
3. **Fit a Bass-type diffusion to the analog set.** Parameterize p (coefficient of innovation) and q (coefficient of imitation) from the analogs, then scale to the new product's addressable base. This gives you a curve with a defensible shape, not a flat line.
4. **Forecast as a range, never a point.** Produce low / base / high scenarios. NPI uncertainty is irreducible before launch — planning the range, and the supply flexibility to cover it, is the only honest response.
5. **Stage the supply commitment to the ramp.** Phase initial builds and pre-commit flexible capacity rather than betting the full launch quantity up front. The staged commitment is what converts forecast uncertainty into a manageable risk instead of a single all-or-nothing bet.
6. **Set a re-forecast trigger on first sell-through.** The first 4–8 weeks of real POS collapse the uncertainty faster than any pre-launch model. Replan hard at that point — the launch forecast was always a placeholder for the data you did not yet have.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A consumer-electronics brand forecast a new SKU by copying its predecessor's flat run-rate — **baseline: ±50% launch error, a week-3 stockout that cost ~$2.1M in undefended-window sales, then a $1.6M write-off of the over-built back half**. It rebuilt with a 4-analog Bass fit, a low/base/high range, staged supply, and a week-4 re-forecast trigger. Results: **launch error ±50% -> ±22%, undefended-window stockout eliminated, write-off $1.6M -> $0.4M (-75%)**.

IN PRACTICE

Weighted analog selection

Score candidate analogs on attribute similarity and weight the launch curve by those scores. The output is a blended ramp drawn from several relevant precedents, with the match logic documented — defensible in an S&OP review in a way a single hand-picked analog never is.

Scenario range, not a point

Always carry low / base / high. The base feeds the plan; the range sizes the supply flexibility and the financial exposure. A single-point NPI forecast is a false precision that drives an all-or-nothing supply bet.

Staged supply commitment

Match the supply commitment to the ramp: an initial build sized to the low/base, with flexible capacity pre-arranged to chase upside. This converts forecast uncertainty into staged, reversible decisions rather than one irreversible build quantity.

First-sell-through re-forecast trigger

Define in advance the data point — typically 4–8 weeks of POS — at which the launch forecast is mandatorily replaced by an actuals-based one. The discipline of a pre-agreed trigger prevents the launch forecast from outliving its usefulness through inertia.

WHAT YOU WOULD USE

NPI / LAUNCH FORECASTING

o9 / SAP IBP — enterprise tier-1 — Lifecycle and analog-based launch forecasting integrated with the planning and supply model, supporting attribute-based analog selection and phase-in profiles. **(o9, SAP — vendor)** The choice when launches must reconcile against staged supply commitments.

Blue Yonder Luminate / Kinaxis — Solid NPI modules with analog/like-modeling and lifecycle curves. **(Blue Yonder, Kinaxis — vendor)** A strong fit where launch cadence is frequent and the analog library is maintained inside the planning platform.

NPI / new-product forecasting specialists — Purpose-built launch-forecasting tools with deep diffusion and conjoint/pre-launch demand methods. **(specialized vendors — vendor)** The vertical option for industries — fashion, consumer tech — where most volume is new each season.

For operations that don't need a platform — A spreadsheet Bass fit on a curated set of analog curves, carried as a low/base/high range with a documented re-forecast trigger. The analog library and the staged commitment discipline deliver most of the value before any specialized tool is purchased.

4 SO WHAT

Operations that forecast launches as a diffusion curve with staged supply defend the window competitors can only enter when you are out of stock; operations that copy a flat predecessor pick which way to lose — give away the share or write off the back half. The flat-line launch forecast is not a forecast; it is a refusal to model adoption. And the launch window is the one moment when being out of stock costs share that does not come back at any markdown.

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